

Insertion anomaly → adding new rows forces user to add data or add empty rows

Deletion anomaly → deleting rows may cause a loss of data

Mod anomaly → changing data in a row forces changes to other rows

Heap → end of file; seq → in order; indexed → seq, faster b tree

Hash → hashing alg: ALL OR NOTHING → no % like only =

Serializability → each trans full processed before another one is allowed to start (uses locks)

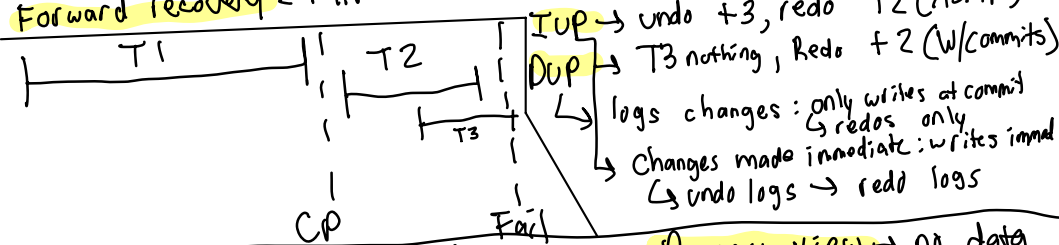
Shared Lock: Read but no update (Just reading a record)

Exclusive Lock: No access permitted (reading and updating)

Versioning → create versions to use with data; commit expires old version; checks versions and handles from there

Backward recovery = rollback (return to earlier state) ×

Forward recovery = rollforward (applies all good transactions) ✓



Create View exp_stuff_v AS
select prod_ID, prodstdprice
from product_T
where prodstdprice > 300
with check option; → insert and set checks

Dynamic view → no data actually stored

Materialized View → copy or replication of data

Dirty read → reading data that hasn't been committed yet, non-repeat → reading same row twice but diff bc someone else updating

Phantom Read → getting diff results bc someone inserted or deleted

Candidate Key → unique, determinat (SSN, IASN), non redundant
→ each field relies on

Deferred: [start] → [Log only] → [commit] → [write]
note: crash before commit = nothing; crash after = redo

Immediate: [start] → [write disk + log] → [commit]
note: crash before commit = undo; crash after = redo

A - Atomicity → "all or nothing"

C - Consistency → "follow the rules"

I - Isolation → "privacy"

D - Durability → "permanence"

Hashing →
Values correspond to only one index so you cannot use % like all values that you search for must be exact

1NF → eliminate MV attributes; select P key
2NF → remove partial func depend: when a non-key depends on part of a primary key

- Create a new relation when the determinat serves as the P key

3NF → remove trans depends; when a non-key is ident by a non-key

- Create new relation for non-key attribute that is a determinat

→ becomes P key

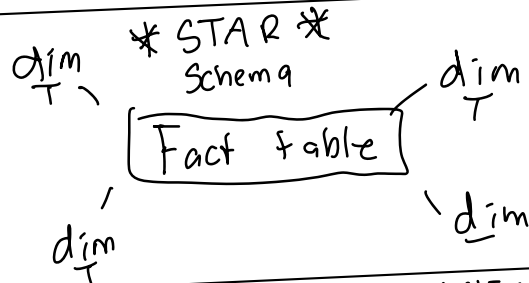
* Func depend: the value of one attribute (the determinat) determines the value of another attribute *

Data WH → a sub oriented, integrated, time variant, non-updatable → for decision making!!

Data Mart → Data WH limited in scope

Dimension tables → contains descriptive data ex: product, customer period of time

Fact tables → contain factual or quantitative data ex: sales, order * 1:N relationship between dim tables and fact tables *



* All relations are in 1NF *

Deadlocks → trans 1 locks A and wants B, while Trans 2 locks B and wants A

Create User bob identified by pass 123

Grant select on customers to bob

* Hints *

each normalization form is after that string has been removed!!!

Fact table → F keys for other tables

SLOW DOWN + THINK

